

Grade 6 Accelerated
Unit 12
Lesson 3 Homework

Name Answer Key
Date _____
Period _____

Solve each problem using proportional reasoning. Show your work.

For questions 1-2, use the following scenario:

A recipe for salad dressing calls for 3 parts oil for every 1 part vinegar.

1. How many tablespoons (tbsp) of vinegar are needed to make $\frac{3}{4}$ cup of salad dressing? 1 cup = 16 tbsp

$$\frac{1 \text{ cup}}{16 \text{ tbsp}} = \frac{\frac{3}{4} \text{ cup}}{12 \text{ tbsp}} \quad \text{oil : vinegar : total} \rightarrow 3 : 1 : 4$$

$$\frac{\text{vinegar}}{\text{total}} \left| \frac{1}{4} = \frac{x}{12 \text{ tbsp}} \right. \quad x = 3 \text{ tbsp of vinegar}$$

2. How many tablespoons of oil are needed to make $\frac{3}{4}$ cup of salad dressing?

$$\frac{\text{oil}}{\text{total}} \left| \frac{3}{4} = \frac{x}{12 \text{ tbsp}} \right. \quad x = 9 \text{ tbsp of oil}$$

3. The retail price of grapes is \$2.10 per pound. If a customer paid \$7.35 for grapes, how many pounds did she buy?

$$\frac{\$}{\text{lb}} \left| \frac{2.10}{1} = \frac{7.35}{x} \right. \quad x = 3.5 \text{ pounds}$$

4. Gregg and Dan are moving boxes. Gregg can move 8 boxes every half hour. Dan can move 11 boxes every half hour. How many hours will it take Gregg and Dan to move all 114 boxes?

$$\frac{\text{boxes}}{\text{hours}} \left| \frac{19}{0.5} = \frac{114}{x} \right. \quad x = 3 \text{ hours}$$

For questions 5-7, use the following scenario:

A recipe for sparkling cranberry pineapple punch calls for 3 cups of pineapple juice, 3 cups of cranberry juice, and 1 quart of ginger ale. Miguel uses to make 7.5 cups of pineapple juice to make punch for his party.

5. How much ginger ale does Miguel need to use for his punch?

$$\begin{array}{c|c} \frac{\text{pineapple}}{\text{ginger ale}} & \frac{3 \text{ cups}}{1 \text{ qt}} = \frac{7.5 \text{ cups}}{x} \end{array} \quad x = 2.5 \text{ quarts}$$

$\xrightarrow{\times 2.5}$
 $\xleftarrow{\times 2.5}$

6. How much cranberry juice does Miguel need to use for his punch?

$$\begin{array}{c|c} \frac{\text{cran}}{\text{pine}} & \frac{3 \text{ cups}}{3 \text{ cups}} = \frac{x}{7.5 \text{ cups}} \end{array} \quad x = 7.5 \text{ cups}$$

$\xrightarrow{\times 2.5}$
 $\xleftarrow{\times 2.5}$

7. By what factor did Miguel scale the recipe up?

2.5 He made 2.5 times the original recipe.

For questions 8-10, use the following scenario:

A teacher is planning a field trip to a theme park. For every 8 students, the teacher must have 1 chaperone.

8. What is the ratio of students to total people on the field trip?

8:9

9. If the teacher purchased 144 total tickets, how many tickets were for students?

$$\begin{array}{c|c} \frac{\text{students}}{\text{total}} & \frac{8}{9} = \frac{x}{144} \end{array} \quad x = 128 \text{ student tickets}$$

$\xrightarrow{\times 16}$
 $\xleftarrow{\times 16}$

10. If the teacher purchased 108 total tickets, how many tickets were for chaperones?

$$\begin{array}{c|c} \frac{\text{chaperones}}{\text{total}} & \frac{1}{9} = \frac{x}{108} \end{array} \quad x = 12 \text{ chaperone tickets}$$

$\xrightarrow{\times 12}$
 $\xleftarrow{\times 12}$